Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	V
Subject Code & Name	ICS1502 -Introduction Machine Learning		
Academic year	2025-2026 (Odd)	Batch: 2023-2028	**Due date: 01/11/2025**

Assignment 1

1 Aim

- To develop and evaluate machine learning models for **Mobile Price Prediction** using regression techniques such as Linear Regression.
- To implement and compare classification algorithms for **Bank Note Authentication**, including Logistic Regression.
- To analyze model performance using appropriate evaluation metrics and visualize results for better interpretability.

2 Libraries Used

Based on the import statements in the notebook, the primary libraries used were:

- **pandas** – For data manipulation, reading CSV files, and data frame operations.
- **numpy** – For numerical computations, especially for array manipulation and mathematical functions.
- **matplotlib** – For creating static, animated, and interactive visualizations.
- **seaborn** – For making attractive and informative statistical graphics.
- **scikit-learn** – For implementing and evaluating machine learning models.

3 Objectives

- To understand and apply supervised machine learning algorithms for regression and classification tasks.
- To perform data preprocessing, feature scaling, and train–test splitting for efficient model training.
- To build predictive models for estimating mobile prices using regression methods such as Linear Regression.
- To classify banknotes as authentic or forged using classification models such as Logistic Regression.
- To evaluate model performance using metrics such as Accuracy, Mean Squared Error (MSE), Confusion Matrix, and Classification Report.
- To visualize the relationship between features and target variables and analyze the effect of outliers and regularization.

4 Mobile Phone Price Prediction (Regression)

Aim

To predict mobile phone prices using linear regression and analyze how closed-form and gradient descent approaches differ in accuracy and computational behavior. The experiment also investigates the effect of L2 regularization and standardization.

Mathematical Intuition

The linear regression model assumes:

$$\hat{y} = X\theta$$

where X is the feature matrix and θ is the parameter vector. The objective is to minimize the Mean Squared Error (MSE):

$$J(\theta) = \frac{1}{2n}\|X\theta - y\|^2$$

Closed-form solution:

$$\theta = (X^T X)^{-1} X^T y$$

Gradient Descent:

$$\theta := \theta - \alpha \frac{1}{n} X^T (X\theta - y)$$

Ridge Regression (L2 Regularization):

$$\theta = (X^T X + \lambda I)^{-1} X^T y$$

introduces a penalty term λ to control weight magnitude and avoid overfitting.

Methodology

1. The dataset was cleaned and divided into 80% training and 20% testing sets.
2. Standardization was applied to ensure numerical stability.
3. The regression model was trained using:
   - Closed-form solution
   - Gradient Descent
   - L2 regularization (Ridge)
4. Predictions were evaluated using RMSE and R^2 score.

Results

Table 1: Performance Comparison of Regression Methods

Method	MSE	R^2 Score
Closed-form (OLS)	31650.99	0.6536
Gradient Descent	25294.28	0.7232
Ridge Regression (Best $\lambda = 100$)	–	0.7622

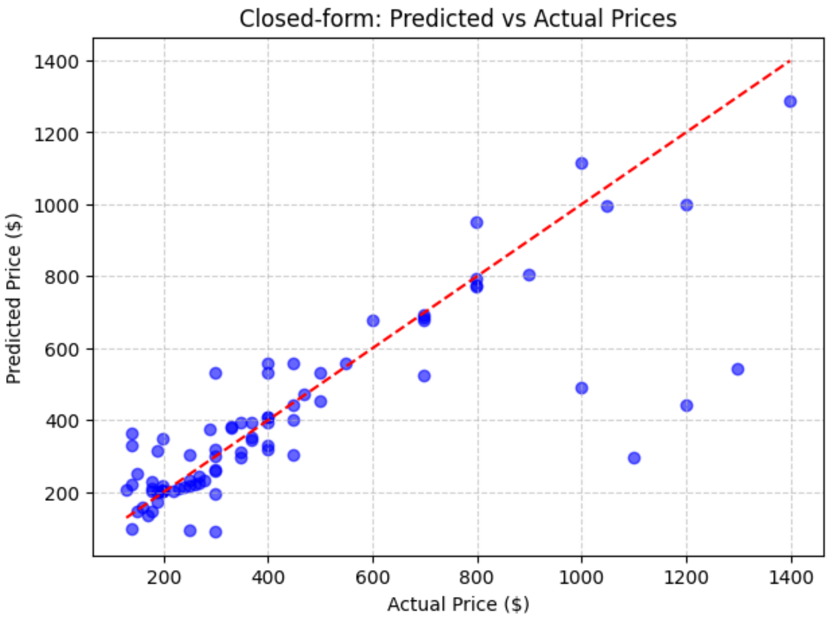

Figure 1: Predicted vs Actual Prices (Closed-form Solution)

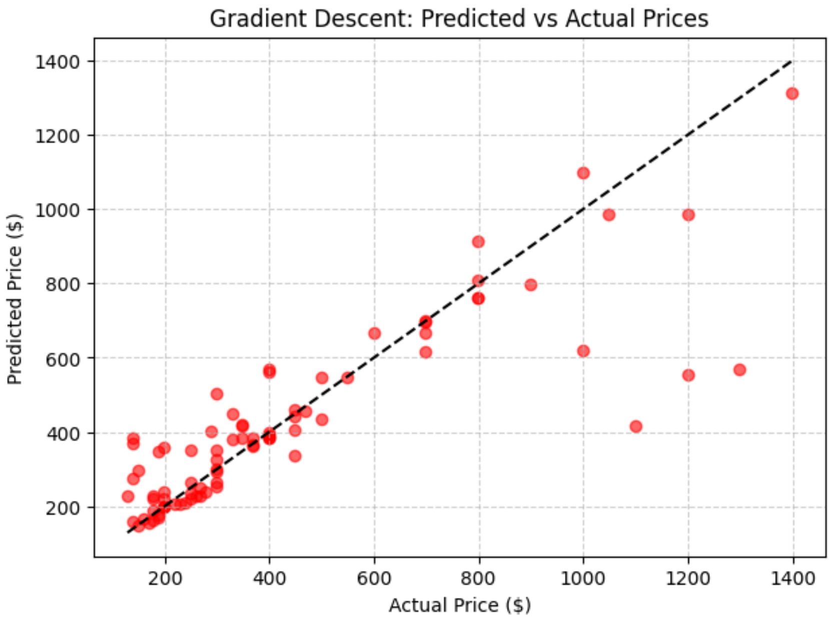

Figure 2: Predicted vs Actual Prices (Gradient Descent)

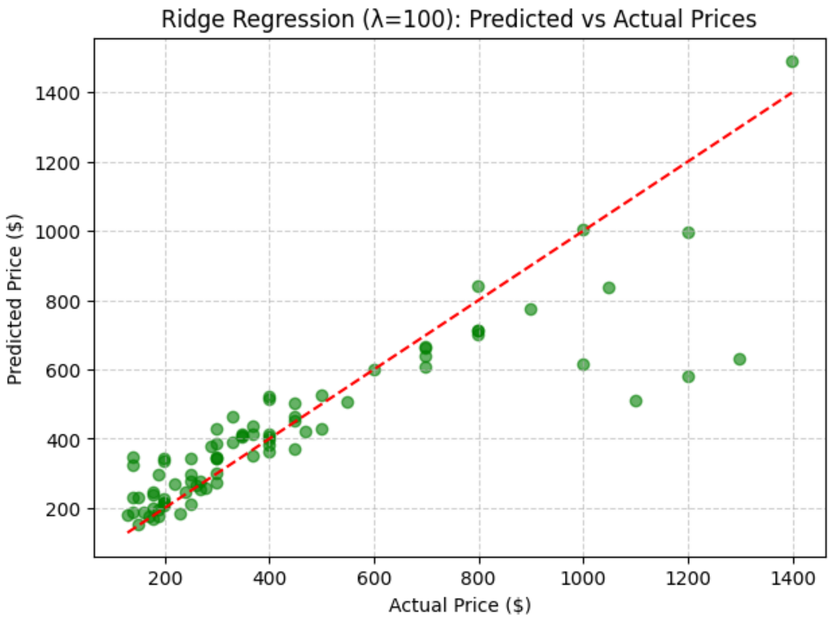

Figure 3: Predicted vs Actual Prices (Ridge Regression)

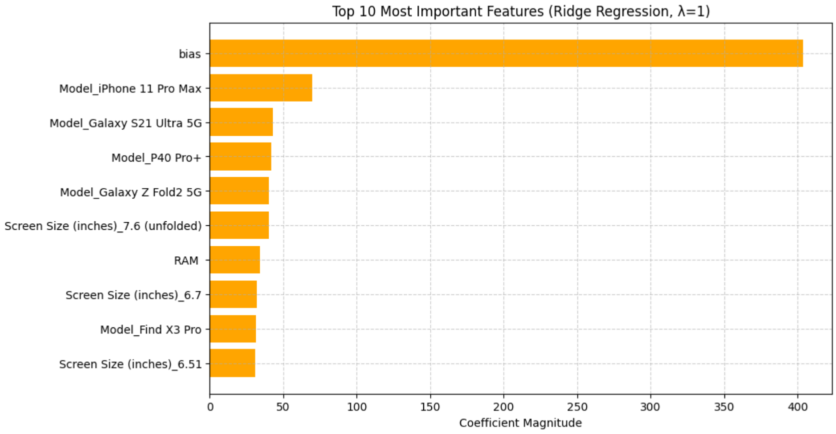

Figure 4: Important Features

Inference

- Standardized Ridge regression achieved the best accuracy ($R^2 = 0.76$).
- Gradient descent converged slower but gave comparable results.
- Closed-form was computationally faster for small data.
- L2 regularization improved generalization and reduced overfitting.

Learning Outcome

- Understood how regression equations can be solved through matrix algebra.
- Observed how L2 regularization affects parameter magnitudes.
- Gained clarity on the influence of data scaling on optimization.

5 Bank Note Authentication (Classification)

Aim

To build a logistic regression classifier that predicts whether a bank note is genuine or forged, and evaluate its performance under regularization and outlier conditions.

Mathematical Intuition

Logistic regression models probability as:

$$P(y = 1|x) = \frac{1}{1 + e^{-X\theta}}$$

Cost function:

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] + \frac{\lambda}{2}\|\theta\|^2$$

Methodology

1. Split dataset into training (80%) and testing (20%).
2. Applied feature standardization.
3. Trained models:
   - Logistic Regression (No Regularization)
   - Logistic Regression (L2 Regularization with varied λ)
4. Evaluated accuracy before and after introducing 10% outliers.
5. Plotted accuracy vs regularization and 3D feature visualization.

Results

Table 2: Performance Comparison of Logistic Regression Models

Model	**Accuracy**
No Regularization	0.975
L2 Regularization ($\lambda = 1$)	0.978
After Outlier Injection	0.961

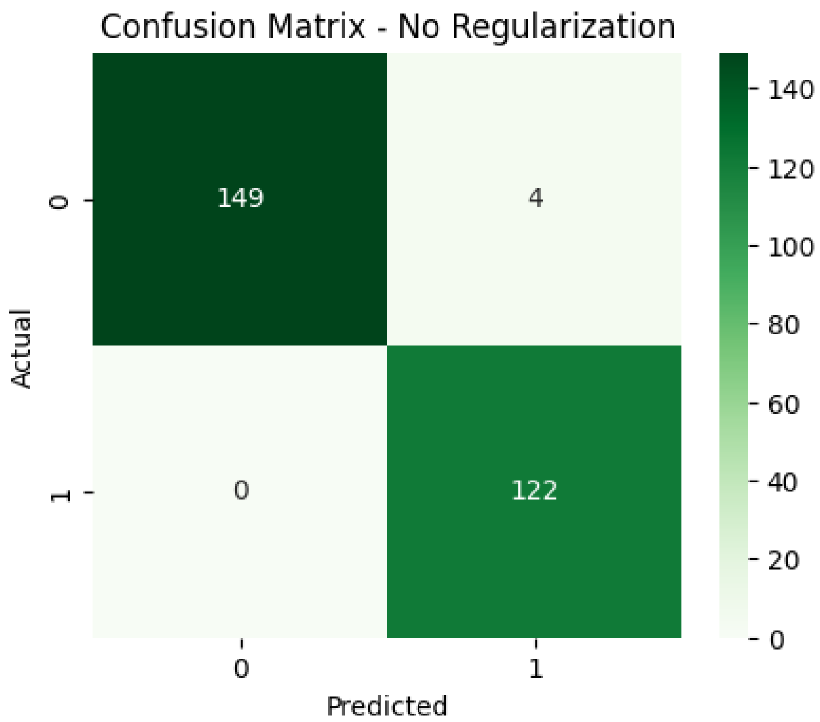

Figure 5: Confusion Matix No Regularization (λ)

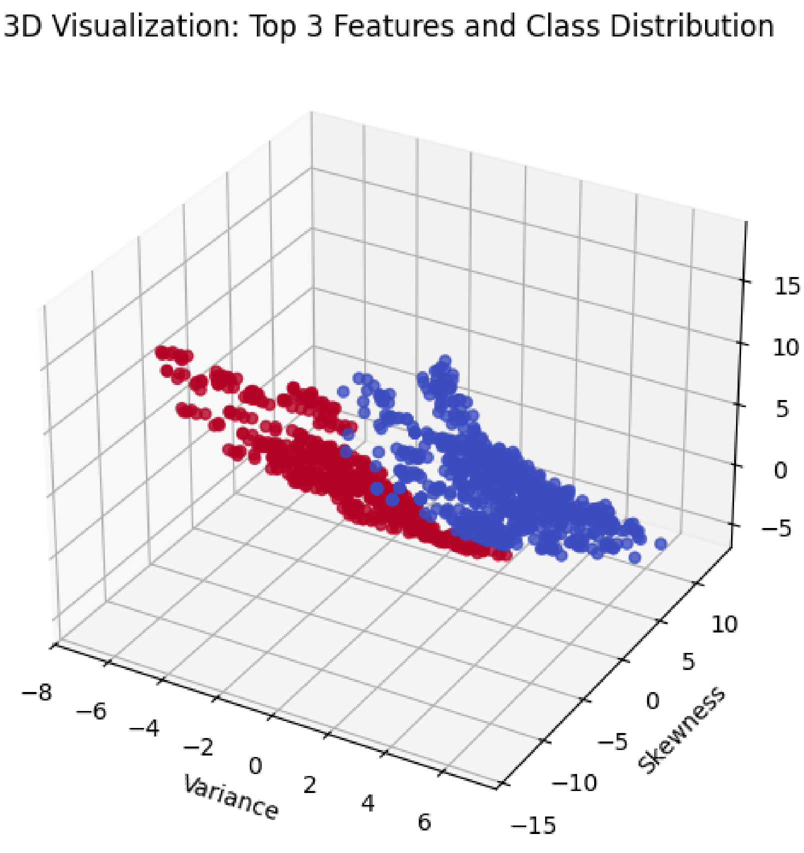

Figure 6: 3D Visualization of Top 3 Important Features

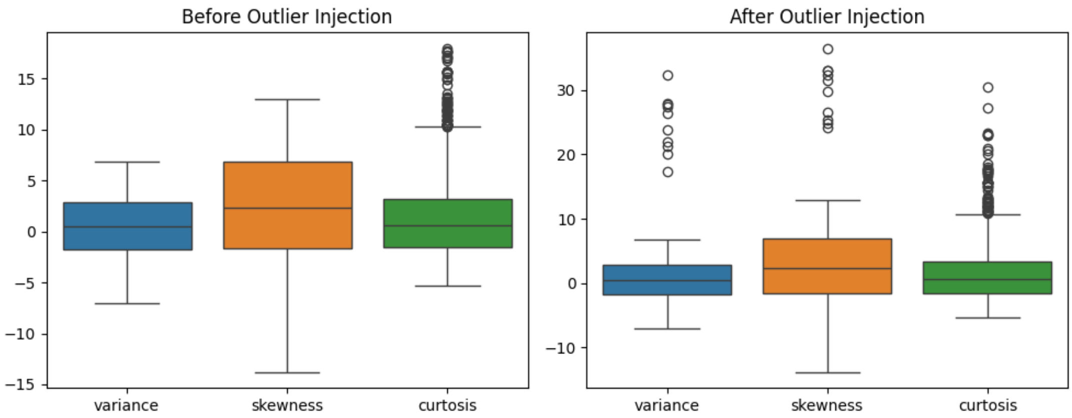

Figure 7: Outliers Before vs After

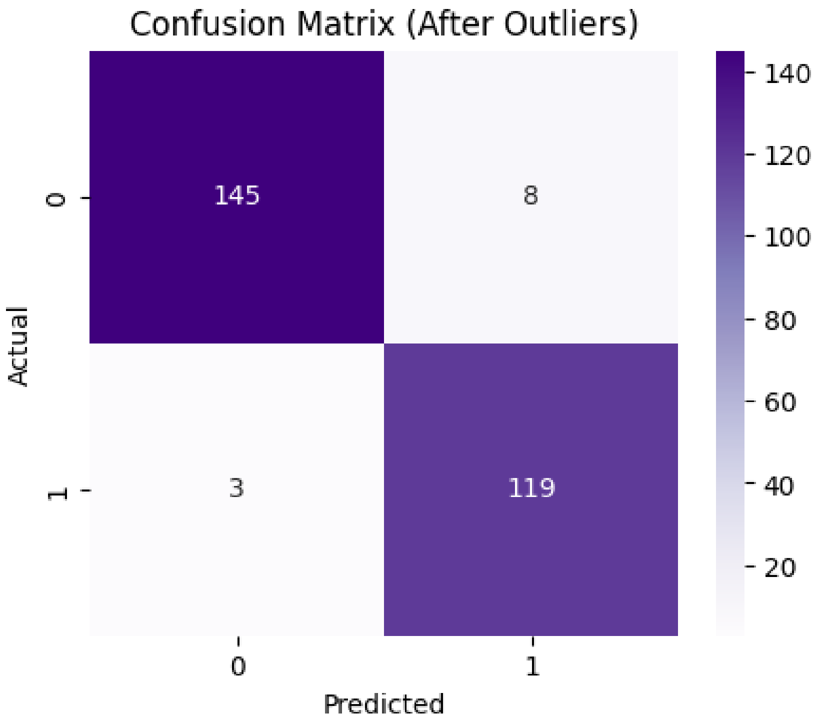

Figure 8: Confusion Matix After Outliers

Inference

- The dataset was linearly separable; logistic regression achieved nearly 98% accuracy.
- L2 regularization improved model robustness.
- Outliers slightly reduced performance, but L2 penalty limited their effect.
- Variance and curtosis were the most important predictive features.

Learning Outcome

- Learned how logistic regression performs under regularization.

- Understood the effect of outliers and penalty term on model stability.
- Visualized decision boundaries and feature importance.

Overall Conclusion

Both regression and classification were implemented successfully using linear models. Regularization and normalization proved vital for performance stability. Linear Regression captured pricing trends, while Logistic Regression efficiently separated genuine and forged notes. The study highlights that mathematical rigor and proper preprocessing ensure robust ML performance.

GitHub Repository

The complete source code for this Assignment is available on GitHub: GitHub